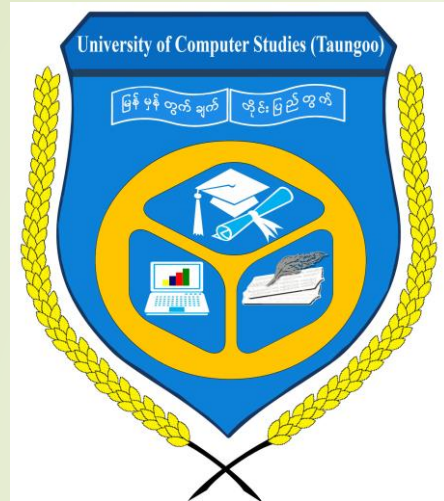




Introduction To Python

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Contents

- SciPy
- Matplotlib

SciPy (Scientific Python)

- **SciPy** is a scientific computation library that use *Numpy* underneath.
- It provides more utility functions for optimization, stats and signal processing.
- Source code: <https://github.com/scipy/scipy>

```
from scipy import constants  
print(constants.liter)           => 0.001  
  
import scipy  
print(scipy.__version__)         => 1.11.4
```

SciPy (Scientific Python)

➤ SciPy Constants

- It provides many built-in *scientific constants* for scientific implementations.

```
from scipy import constants  
print(constants.giga) =>1000000000.0           # Metric (SI) Prefixes  
print(constants.milli) =>0.001  
print(constants.kibi) =>1024                   # Binary Prefixes  
print(constants.mebi) =>1048576  
print(constants.gram) =>0.001                  # Mass  
print(constants.atomic_mass) =>1.66053904e-27  
print(constants.degree) =>0.017453292519943295 # Angle  
print(constants.arcmin) =>0.0002908882086657216  
print(constants.minute) =>60.0                 # Time  
print(constants.hour) =>36000
```

SciPy (Scientific Python)

► SciPy Constants

```
from scipy import constants
```

```
print(constants.inch) =>0.0254
```

Length

```
print(constants.zero_Celsius) =>273.15
```

Temperature

```
print(constants.atmosphere) =>101325.0
```

Pressure

```
print(constants.mmHg) =>133.32236842105363
```

```
print(constants.hectare) =>10000.0
```

Area

```
print(constants.electron_volt) =>1.6021766208e-19
```

Energy

```
print(constants.liter) =>0.001
```

Volume

```
print(constants.horsepower) =>745.6998715822701
```

Power

```
print(constants.speed_of_sound) =>340.5
```

Speed

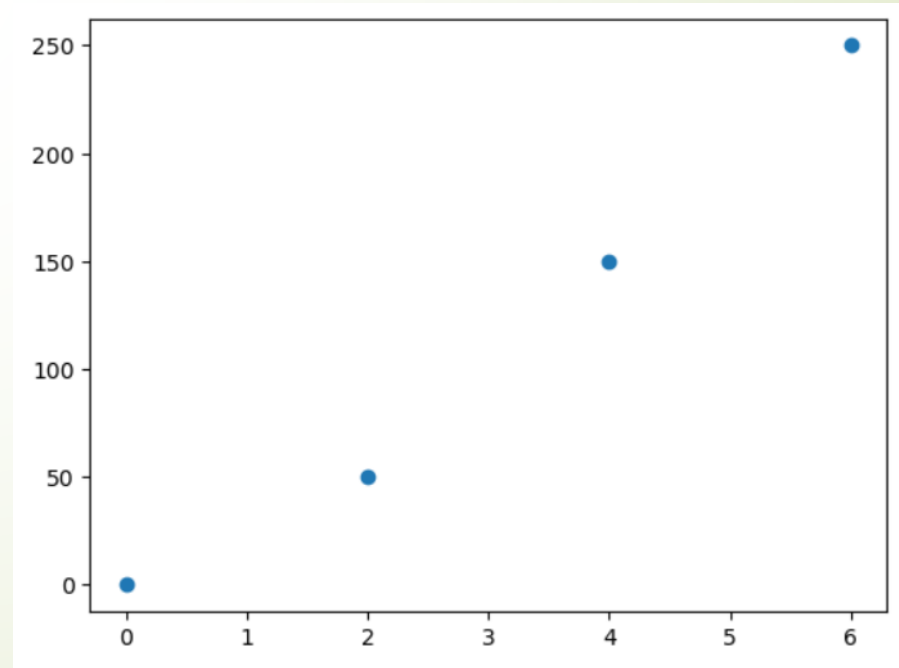
```
print(constants.kilogram_force) =>9.80665
```

Force

Python Matplotlib

- **Matplotlib** is a low level graph plotting library that serves as visualization utility.
- It was created by John D. Hunter and open source.
- Source code: <https://github.com/matplotlib/matplotlib>
- **Matplotlib Pyplot**
- Most of the Matplotlib utilities lies under the *pyplot* submodule.

```
import matplotlib.pyplot as plt
import numpy as np
xaxis= np.array([0,2,4,6])
yaxis = np.array([0,50,150,250])
plt.plot(xaxis, yaxis, 'o')
plt.show( )
```

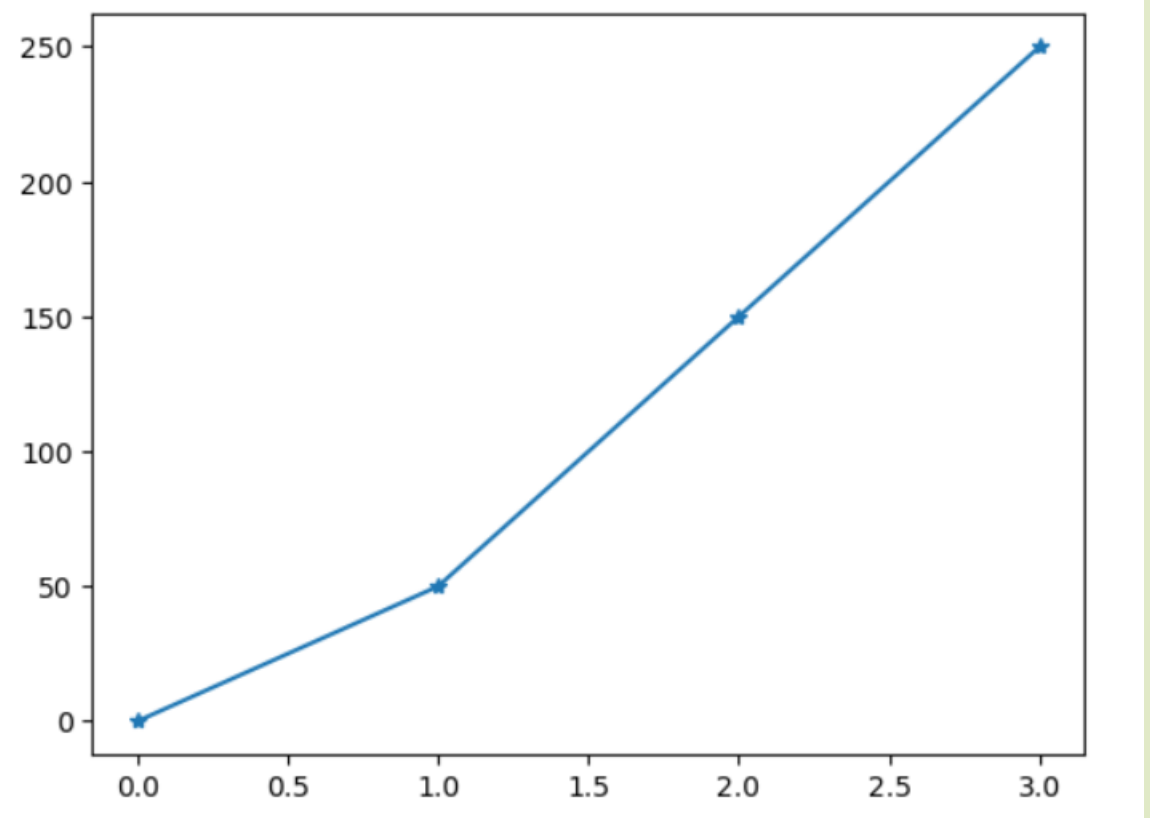


Python Matplotlib

➤ Matplotlib Markers

- To emphasize each point with a specified marker, *marker keyword argument* can be used.

```
import matplotlib.pyplot as plt
import numpy as np
yaxis = np.array([0,50,150,250])
plt.plot(yaxis, marker = '*')
plt.show( )
```



Python Matplotlib

➤ Matplotlib Line

- To change the style of the plotted line, *linestyle* or *ls* keyword argument can be used.
- To set the color of the line, *color* or *c* keyword argument can be used.
- To change the width of the line, *linewidth* or *lw* keyword argument can be used.

```
import matplotlib.pyplot as plt
import numpy as np
yaxis = np.array([0,50,150,250])
plt.plot(yaxis, linestyle = 'dotted',
marker = '*', color = 'r', lw = '20.5')
plt.show( )
```

Style	or
'solid'(default)	'-'
'dotted'	'.'
'dashed'	'--'
'dashdot'	'-.'

```
import matplotlib.pyplot as plt
import numpy as np
x1=np.array([0,1,2,3])
y1=np.array([3,8,1,10])
x2=np.array([0,1,2,3])
y2=np.array([6,2,7,11])
plt.plot(x1,y1,x2,y2)
plt.show( )
```

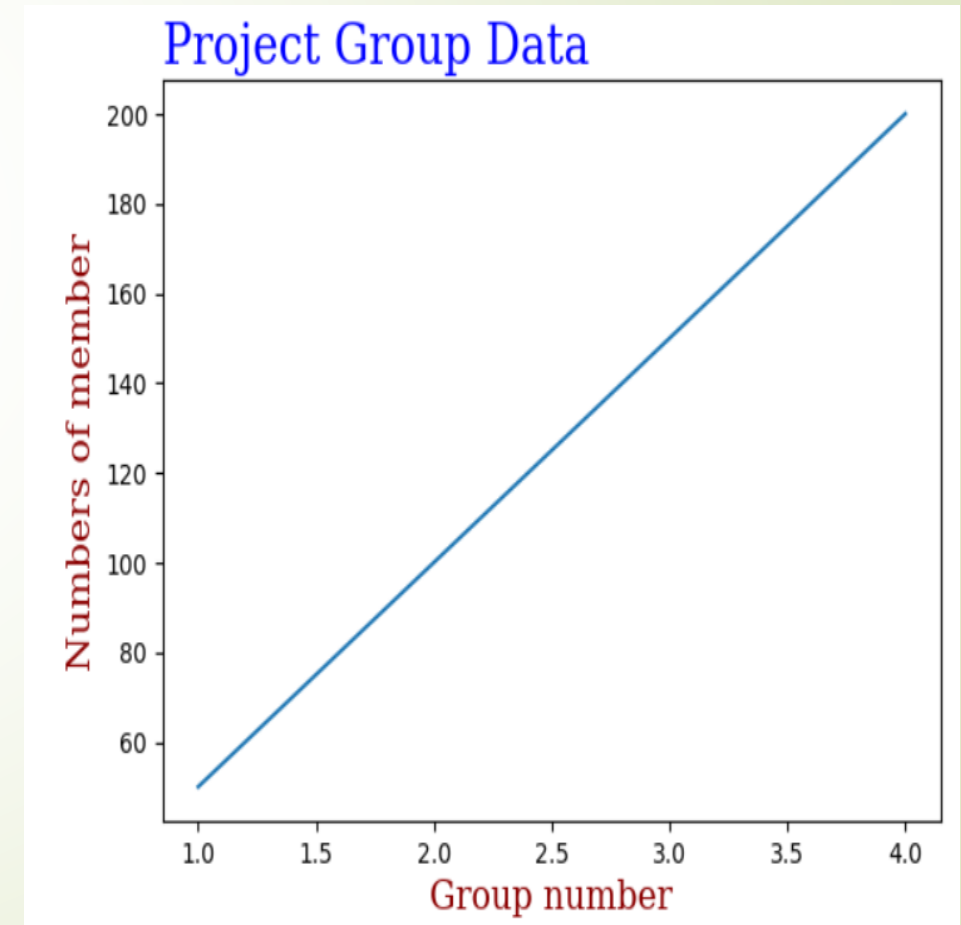
Python Matplotlib

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➤ Matplotlib Labels and Title

- To set a label for x and y axis, *xlabel()* and *ylabel()* functions can be used.
- To set a title for the plot, *title()* function can be used.

```
import matplotlib.pyplot as plt
import numpy as np
xaxis = np.array([1,2,3,4])
yaxis = np.array([50,100,150,200])
f1={'family':'serif', 'color':'blue', 'size':20}
f2={'family':'serif', 'color':'darkred', 'size':15}
plt.title("Project Group Data", fontdict = f1, loc='left')
plt.xlabel("Group number", fontdict = f2)
plt.ylabel("Numbers of member ", fontdict = f2)
plt.plot(xaxis, yaxis)
plt.show( )
```



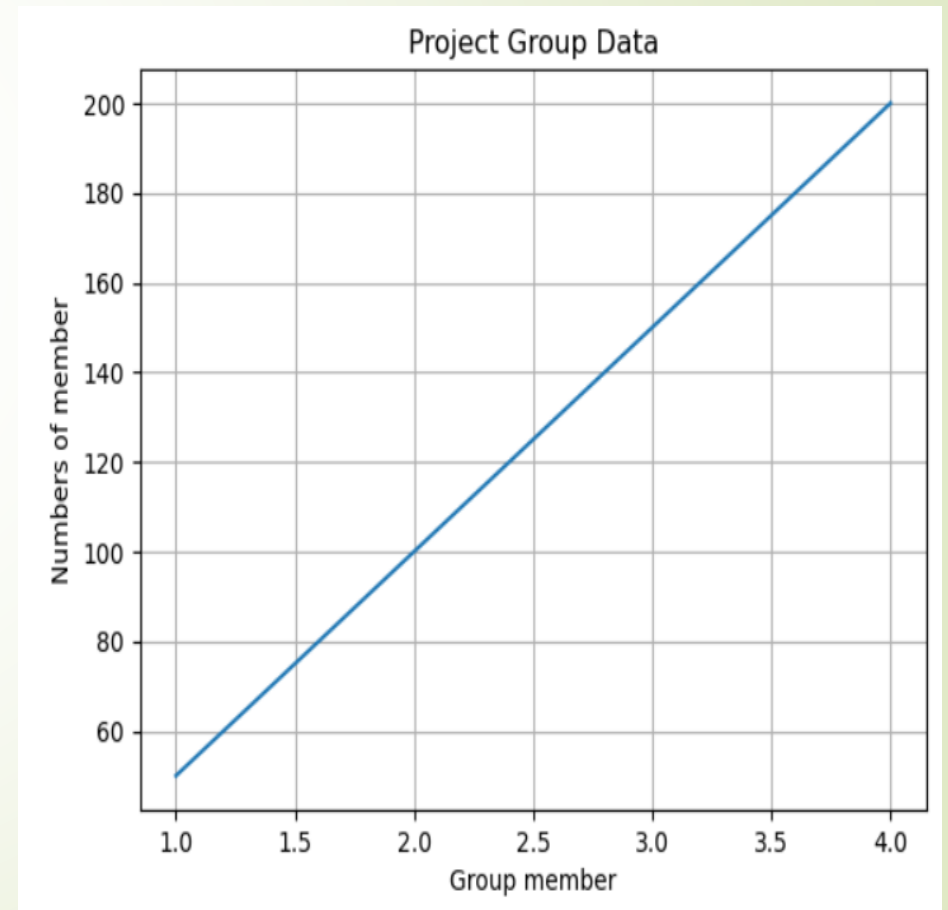
Python Matplotlib

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➤ Matplotlib Grid Lines

➤ To add grid line to the plot, *grid()* function can be used.

```
import matplotlib.pyplot as plt
import numpy as np
xaxis = np.array([1,2,3,4])
yaxis = np.array([50,100,150,200])
plt.title("Project Group Data")
plt.xlabel("Group number")
plt.ylabel("Numbers of member ")
plt.plot(xaxis, yaxis)
plt.grid( ) # plt.grid(axis='x') # plt.grid(axis='y')
plt.show( )
```



Python Matplotlib

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➤ Matplotlib Subplot

➤ To draw multiple plots in one figure, *subplot(row, col, index)* function can be used.

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
x = np.array([0,1,2,3])
```

```
y = np.array([3,8,1,10])
```

```
plt.subplot(1, 2, 1)
```

```
plt.plot(x,y)
```

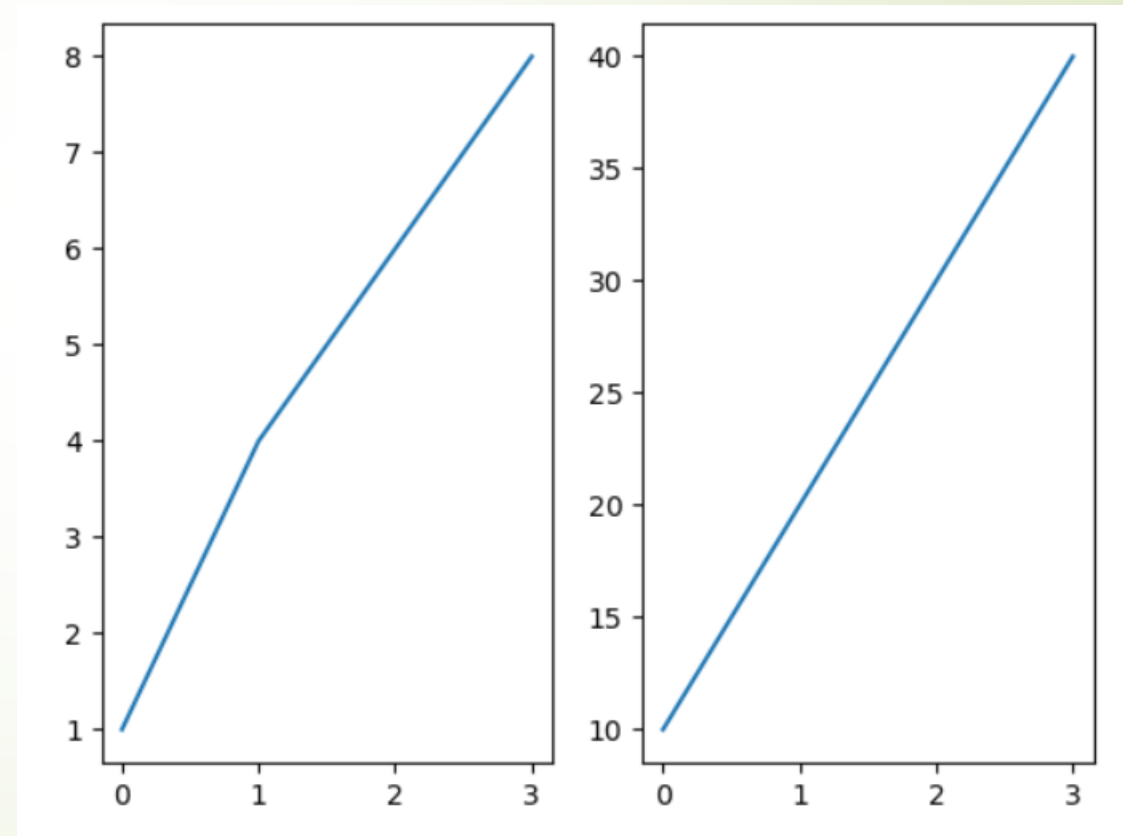
```
x=np.array([0,1,2,3])
```

```
y=np.array([10,20,30,40])
```

```
plt.subplot(1, 2, 2)
```

```
plt.plot(x,y)
```

```
plt.show( )
```



Python Matplotlib

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➤ Matplotlib Subplot

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
x = np.array([0,1,2,3])
```

```
y = np.array([3,8,1,10])
```

```
plt.subplot(2,3,1)
```

```
plt.plot(x,y)
```

```
x=np.array([0,1,2,3])
```

```
y=np.array([10,20,30,40])
```

```
plt.subplot(2,3,2)
```

```
plt.plot(x,y)
```

```
x = np.array([0,1,2,3])
```

```
y = np.array([3,8,1,10])
```

```
plt.subplot(2,3,3)
```

```
plt.plot(x,y)
```

```
x=np.array([0,1,2,3])
```

```
y=np.array([10,20,30,40])
```

```
plt.subplot(2,3,4)
```

```
plt.plot(x,y)
```

```
x = np.array([0,1,2,3])
```

```
y = np.array([3,8,1,10])
```

```
plt.subplot(2,3,5)
```

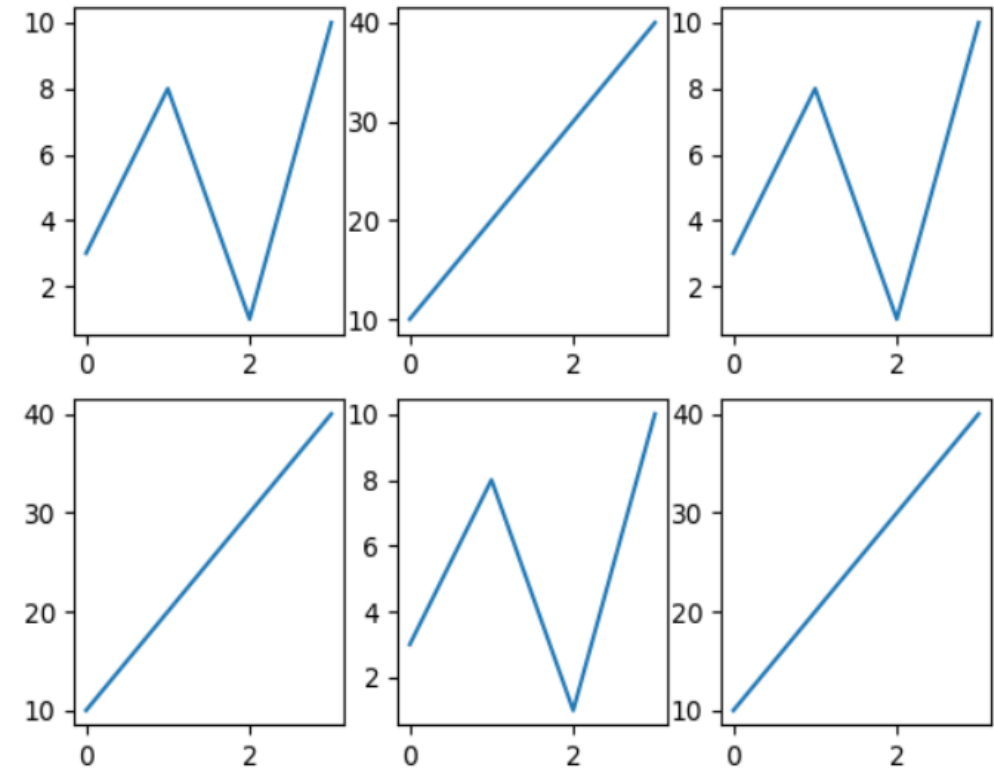
```
plt.plot(x,y)
```

```
x=np.array([0,1,2,3])
```

```
y=np.array([10,20,30,40])
```

```
plt.subplot(2,3,6)
```

```
plt.plot(x,y)
```



Python Matplotlib

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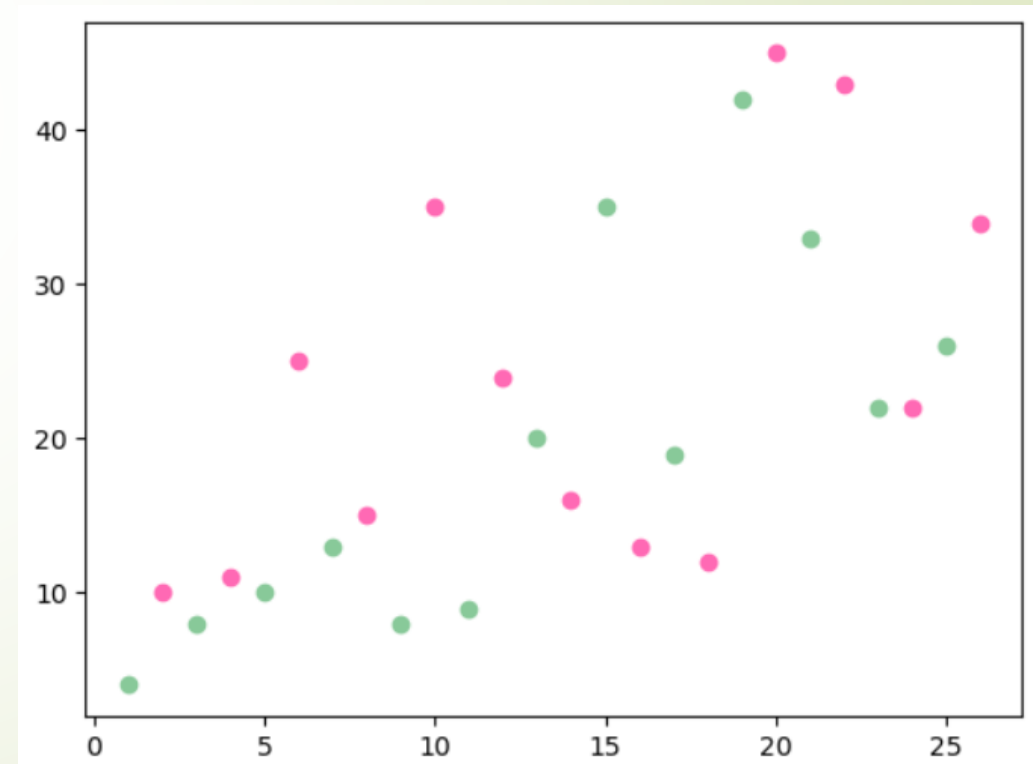
➤ Matplotlib Scatter Plots

- To draw a scatter plot, `scatter( )` function can be used.
- This function plots one dot for each observation.
- It needs two arrays of the same length (**`length(x)=length(y)`**).

```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([2,4,6,8,10,12,14,16,18,20,22,24,26])
y = np.array([10,11,25,15,35,24,16,13,12,45,43,22,34])
plt.scatter(x,y, color='hotpink')

x = np.array([1,3,5,7,9,11,13,15,17,19,21,23,25])
y = np.array([4,8,10,13,8,9,20,35,19,42,33,22,26])
plt.scatter(x,y, color='#88c999')
plt.show( )
```



Python Matplotlib

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➤ Matplotlib Bars

➤ To draw bar graphs, *bar()* can be used.

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

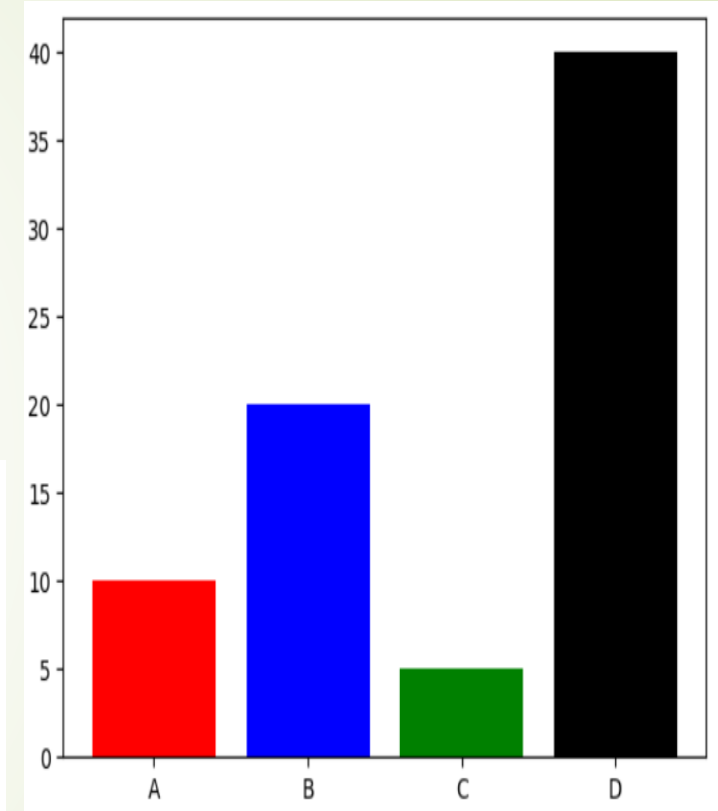
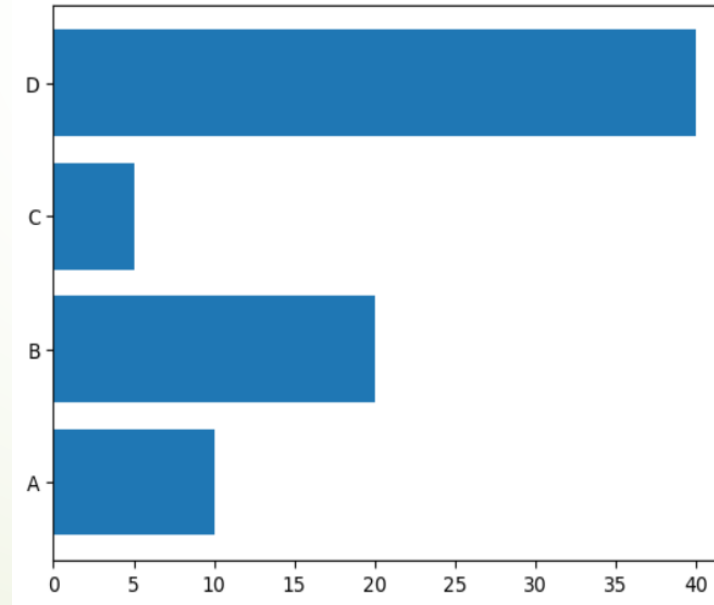
```
x = np.array(["A", "B", "C", "D"])
```

```
y = np.array([10, 20, 5, 40])
```

```
c = np.array(['red', 'blue', 'green', 'black'])
```

```
plt.bar(x,y, color=c)    #plt.barh(x,y)
```

```
plt.show( )
```



Python Matplotlib

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➤ Matplotlib Pie Charts

- To draw pie charts, *pie()* function can be used.

```
import matplotlib.pyplot as plt
```

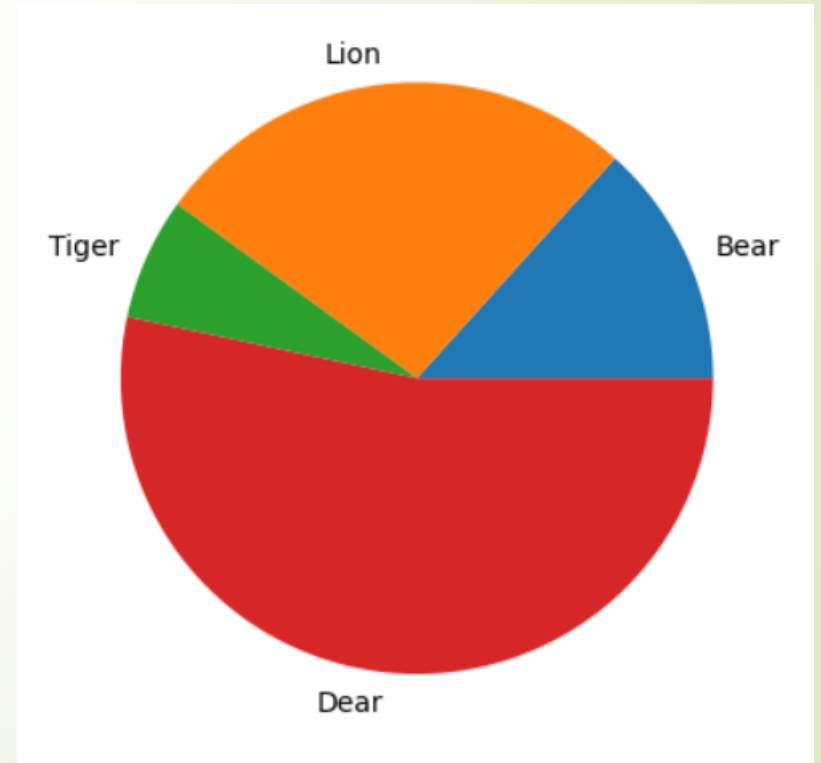
```
import numpy as np
```

```
y = np.array([10, 20, 5, 40])
```

```
mylabels=['Bear', 'Lion', 'Tiger', 'Deer']
```

```
plt.pie(y, labels=mylabels)
```

```
plt.show( )
```



Exercises

- <https://pynative.com/python-operators-and-expression-quiz/>
- <https://pynative.com/python-variables-and-data-types-quiz/>
- <https://pynative.com/basic-python-quiz-for-beginners/>
- <https://pynative.com/python/quizzes/>
- <https://pynative.com/python-basic-exercise-for-beginners/>
- https://www.w3schools.com/python/python_functions.asp

Thank You